Solutions and teaching notes, chapter 1

Teacher use. The following answers concern the tasks not solved in the lesson. Do first make the approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 11 : multiplication comes before addition. *Action:* Rewrite the steps of two calculations with parentheses.
2. $1,\ 8,\ \frac{1}{4}$. *Action:* Build the chain of powers by dividing by 2.
3. $80 \times 1{,}05 = 84$. *Action:* Match 5 % with $\frac{5}{100}$ then by the coefficient 1.05.
4. -2 and 10. *Action:* Replace x in parentheses before calculating.
5. $10 - 3 + 1 = 8$. *Action:* Write out the list then recover the calculation.
6. $0, 1, 2, 3$. *Action:* Draw a two-column table, index and value; test range(1,4).

Investigation

Offer A: last 45 kg, total $\frac{6(20+45)}{2}$ = 195 kg. B: $\frac{20(1-1{,}2^6)}{1-1{,}2} = 198{,}5984$ kg. B exceeds A in total, even if some initial terms are smaller. Calculator use is allowed here; argue about the running total and not only about the last term.

Direct applications

1. $2, 3, 6, 11$. The formula $w_n = n^2$ +2 works: it gives the first term and ($n+1)^2+2-(n^2+2) = 2n+1$.
2. $37 - 17 = 5r$, therefore r = 4 then $u_0 = 17 - 3 \times 4 = 5$.
3. $u_{n+1} - u_n = -4 \times 3^n < 0$: decreasing. v_n alternates between 2 and -2: neither increasing nor decreasing.
4. 4 + 8 + 16 + 32 = 60, or $\frac{4(1-2^4)}{1-2} = 60$.
5. u_3 = 6 : index 3 for ≤ 6, index 4 for < 6.

Exercises: methods and answers

Exercise 1.1 · Getting to grips with indices. $u_0 = 0, u_1 = -1, u_4$ = 20. Expand with parentheses: $u_{n+1} = 2(n+1)^2 - 3(n+1) = 2n^2 + n - 1$; difference 4 $n - 1$. The sign changes between indices 0 and 1: the sequence first decreases, then increases from 1.

Exercise 1.2 · Choose the right model. $a_n = 100 + 15n$ and $b_n = 100(1{,}1)^n$, therefore $a_2 = 130$, b_2 = 121. A assumes a constant net addition; B a constant net rate. In reality, rounding and the carrying capacity limit these models.

Exercise 1.3 · Recovering a sequence. $u_3 = q^2 u_1$ so $q^2 = 9$. Positivity requires $q = 3$, hence $u_0 = 2$ and $u_5 = 486$. Without this detail, $q = -3$ would also be possible, with $u_0 = -2$ and the same $u_5 = 486$.

Exercise 1.4 · Rows of seats. The last row has $18 + 11 \times 2 = 40$ seats. Sum $\frac{12(18+40)}{2} = 348$. Check: a mean of 29 seats per row, between 18 and 40.

Exercise 1.5 · Adding up squares. We get 4 then 9 squares. A square with integer side n completed with a row and a column requires $2n + 1$ squares. The formula n^2 has the right first term and $(n+1)^2 - n^2 = 2n + 1$. The generation is the same.

Exercise 1.6 · Two repayment plans. For A: $a_n = 600 - 50n$ for $0 \leq n \leq 12$, then 0. For B: $b_n = \frac{600}{2^n}$. $b_5 = 18{,}75$ and $b_6 = 9{,}375$; strict threshold at index 6. B never gives exactly 0 in this ideal model: do not claim a finite repayment.

Exercise 1.7 · Reading a list. The first three elements are 2, 5, 14. An initial value and five appends give six elements; indices 0 to 5. The last one is $u_5 = 365$.

Exercise 1.8 · Delayed review. $u_7 = 12 + 5(-3) = -3$. Six terms: $\frac{6(12-3)}{2} = 27$. The 5 counts the loop iterations, the 6 the terms.

CQFD, exercise 9 p. 64, questions a and b. a : $-1, 0, 3, 8$. b : $2, \frac{7}{2}, 4, \frac{17}{4}$. The first index is 0 ; the denominator of b is $n + 1$ and does not vanish on $\mathbb{N}$.

CQFD, exercise 11 p. 64, all questions. From $u_0 = 2$, the rule $u_{n+1} = 2u_n - 1$ gives $u_1 = 3$, $u_2 = 5$, $u_3 = 9$, $u_4 = 17$. The rule uses the previous term, not its index.

MCQ: answers and reasons

MCQ 1.

1. A. $3 \times 4 + 2 = 14$.
2. C. Three additions: $2 + 3 \times 5 = 17$.
3. D. The coefficient is $1 + \frac{8}{100}$.
4. B. $9 - 3 + 1 = 7$.
5. C. $\frac{20 \times 21}{2} = 210$.

MCQ 2.

1. D. The difference is $-3 \times 2^n < 0$.
2. A. Four terms; $\frac{1 - 2^4}{1 - 2} = 15$.
3. C. $u_3 = 2$, $u_4 = 1$.

4. B. The terms alternate between two distinct values.

5. D. Eleven indices from 0 to 10.

True or false

1. False: $u_n = -2^n$ is strictly decreasing.

2. False: $u_n = 0$ and $v_n = n(n-1)(n-2)$ coincide at indices 0, 1, 2, but not at index 3.

3. True: each term is 1 and there are n+1 terms.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language mastered, then require a precise mathematical sentence in French.